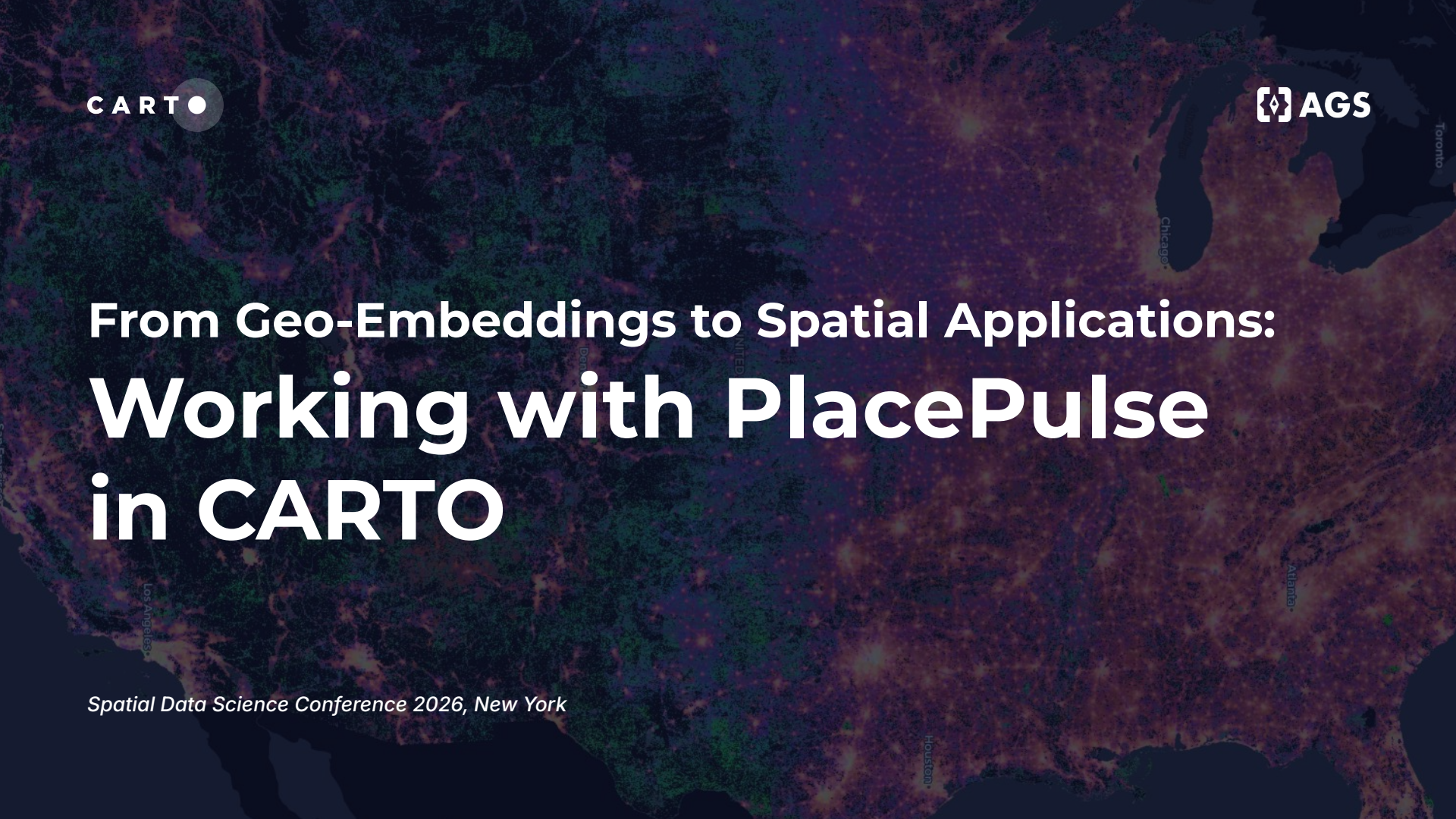


From Geo-Embeddings to Spatial Applications:

Working with PlacePulse in CARTO

Spatial Data Science Conference 2026, New York

Introductions



Lucía García - Duarte

Senior Data Scientist

CARTO

/luciagds/



Gary Menger

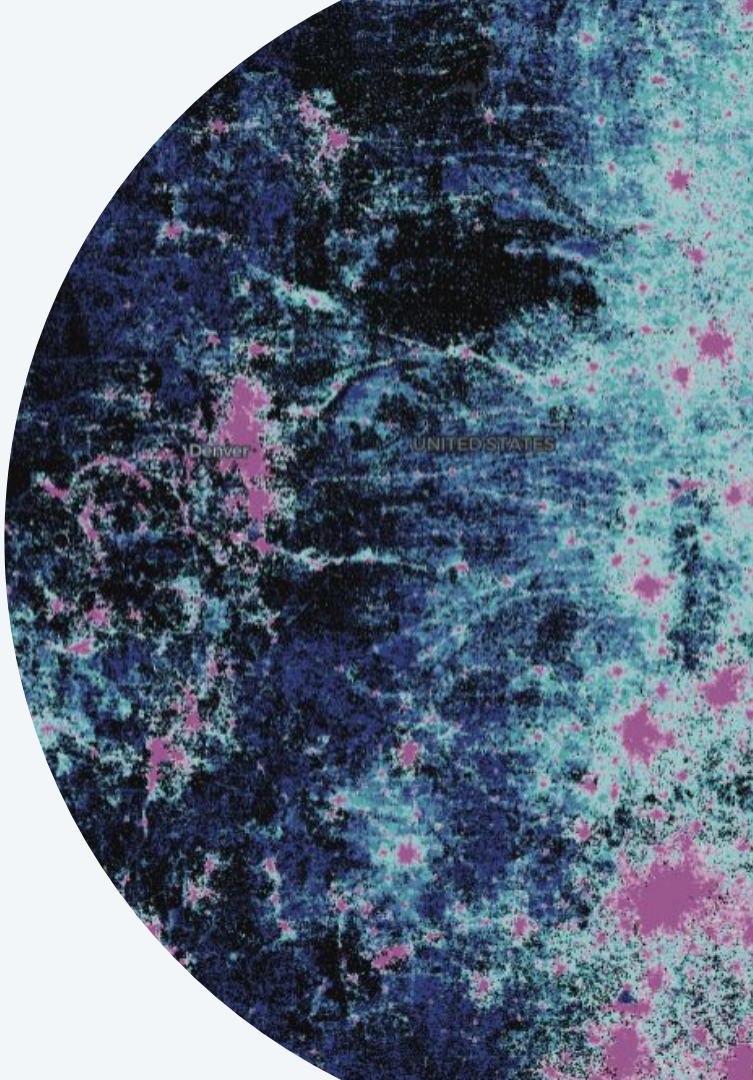
President

Applied Geographic Solutions

/gary-menger-61658513/

Agenda

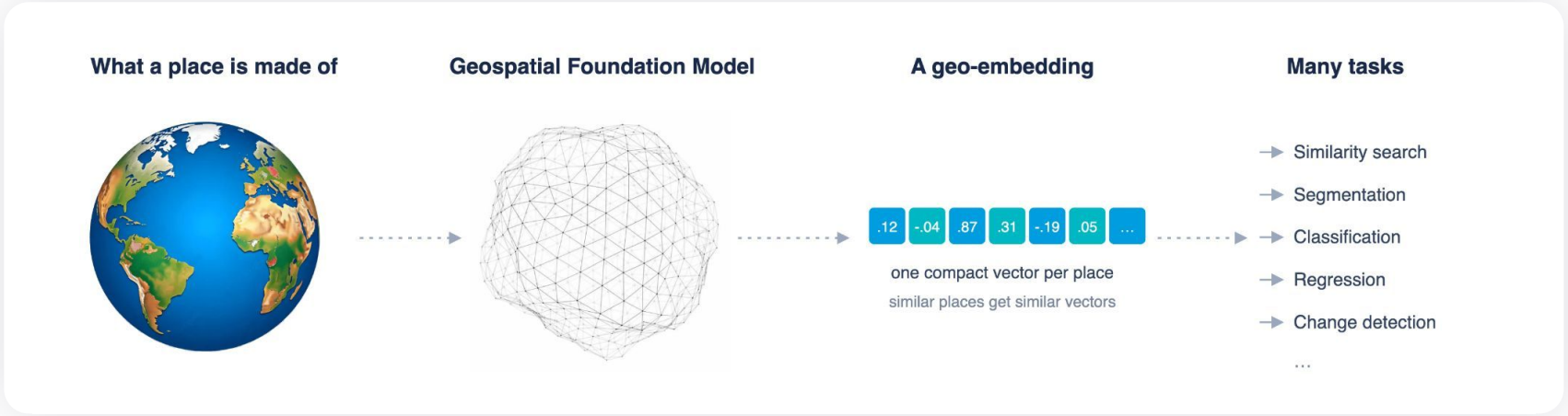
- **Geospatial Foundation Models**
  A quick view on how they are built and what is their purpose
- **Introducing PlacePulse Embeddings**
  How PlacePulse embeddings are built and used in spatial data pipelines
- **The data used to train the embeddings**
  The Applied Geographic Solutions data product
- **Presenting the use case**
  Business use case
- **Hands-on session**
  The live workshop session



01

A quick view on Geospatial Foundation Models

Geospatial Foundation Models



Large, general-purpose AI models trained on massive, multimodal geospatial data (imagery, POIs, mobility, online behavior...)

Learn a unified representation of place through embeddings that summarize the full context of any location

Adaptable to many downstream tasks, from change detection to similarity search, or predictive modeling

PlacePulse Embeddings

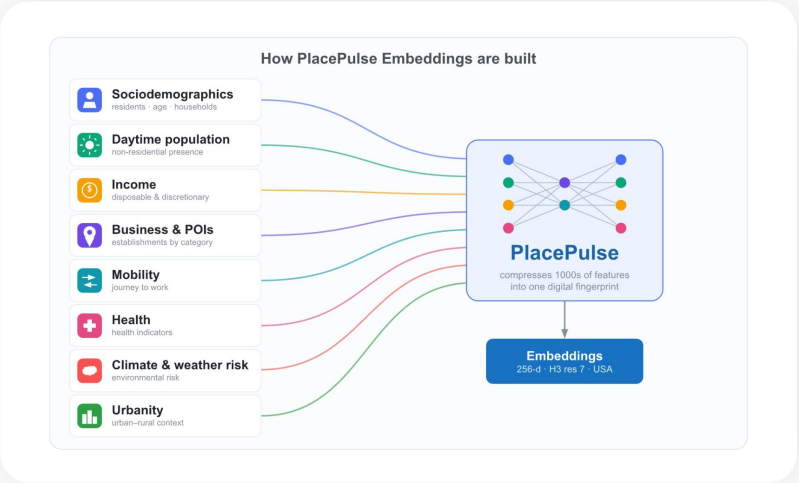
GEOID	emb_0	emb_1	emb_2	emb_3	250 more	emb_254	emb_255
870c00042fffffff	−0.00252	+0.06192	+0.00643	+0.06363	…	−0.04647	−0.01941
870c00050fffffff	−0.00692	+0.07634	+0.01386	+0.08080	…	−0.02664	+0.00749
870c00051fffffff	−0.00153	+0.02937	+0.02129	+0.07387	…	−0.06270	−0.02881
870c00053fffffff	−0.00318	+0.05508	+0.00436	+0.07769	…	−0.05476	−0.01028
870c00054fffffff	+0.00236	+0.09534	−0.01839	+0.04120	…	−0.05367	+0.01604
870c00055fffffff	+0.00277	+0.03738	+0.02358	+0.09321	…	−0.04422	−0.01574
870c0005cfffffff	−0.00624	+0.06570	−0.00585	+0.07253	…	−0.04590	−0.00388
870c000c5fffffff	+0.01177	+0.06980	+0.00145	+0.06041	…	−0.03982	−0.01694
870c000edfffffff	+0.01499	+0.07002	−0.00860	+0.04914	…	−0.04119	−0.01967
870c0019dfffffff	+0.03214	+0.12984	+0.00940	+0.00952	…	+0.03078	−0.01692

02

What are PlacePulse Embeddings



Built from the structural DNA of a place



8 structural themes (2025)

Demographics, business, income, mobility, health, climate risk and more.

2,288 attributes compressed

The cell's own attributes plus a summary of its neighbourhood.

256-d vector per cell

One compact fingerprint on an H3 resolution-7 grid.

~1 million H3r7 cells, full US

Complete and consistent coverage nationwide.

How the model is built

An autoencoder foundation model

Compresses 2,288 standardized attributes into one 256-dimension vector.

Self-supervised, no labels

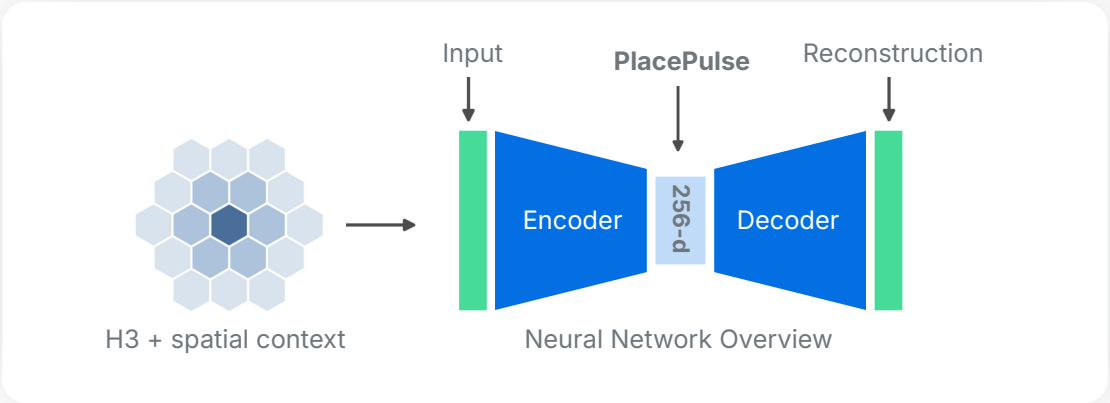
Trained to reconstruct its inputs, so the compact form keeps what matters.

Spatially aware embeddings

Apart from each own H3 cell's features, includes additional signal from neighbouring cells.

Profile, not size

Inputs are standardized and vectors are unit-length, population is factored out.



A fingerprint for every place

A digital fingerprint of place

A compact vector that captures each area's overall character.

Similar places get similar vectors

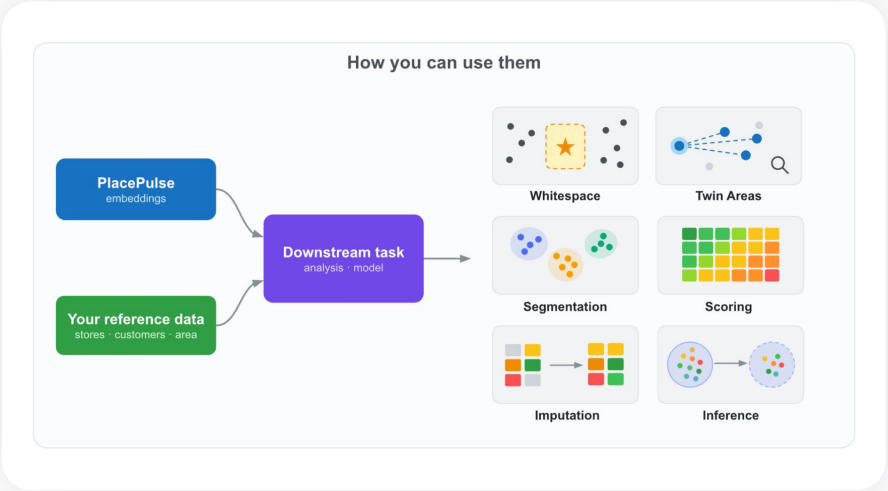
Places that differ end up farther apart in the vector space.

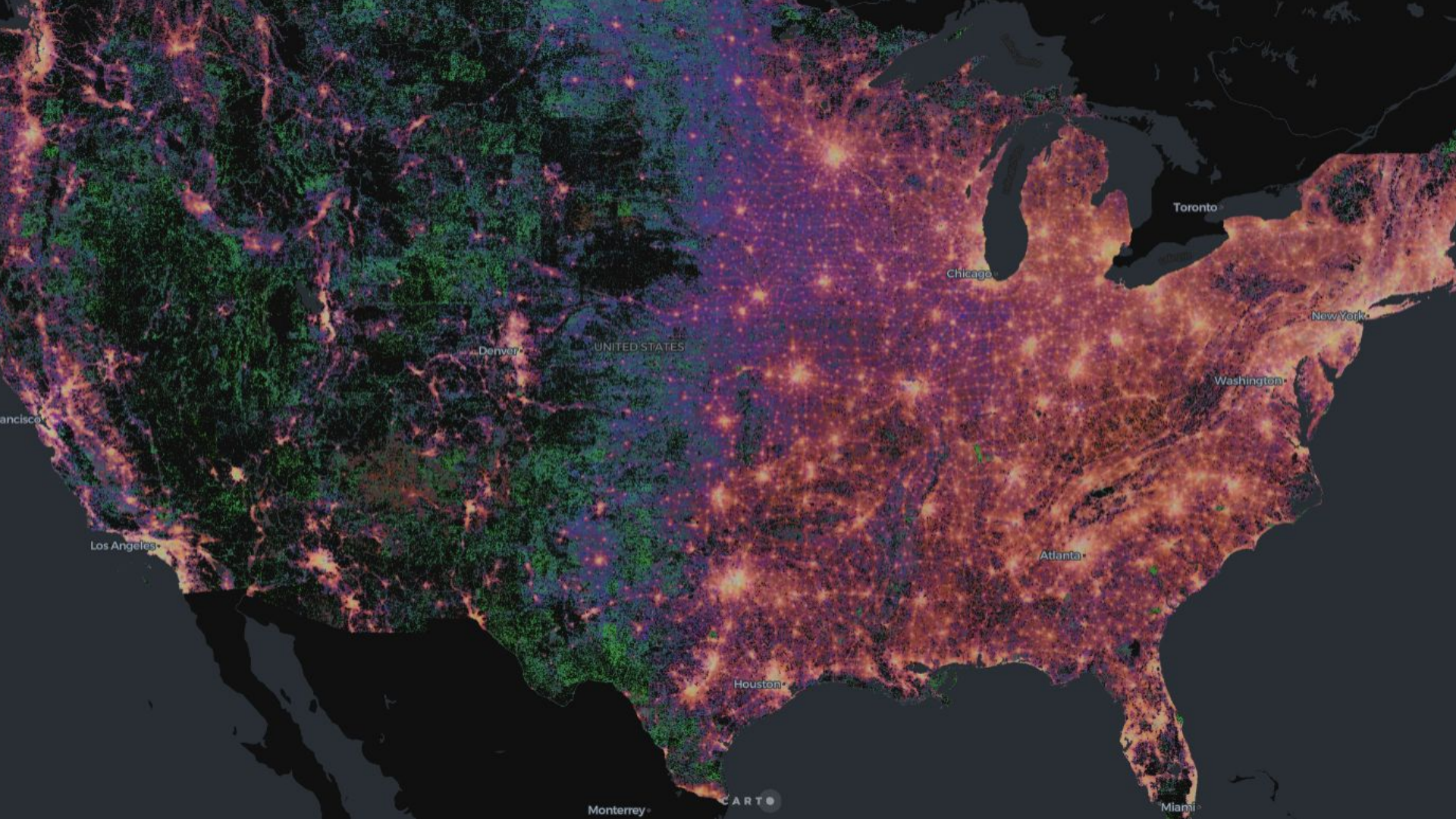
Complete, consistent US coverage

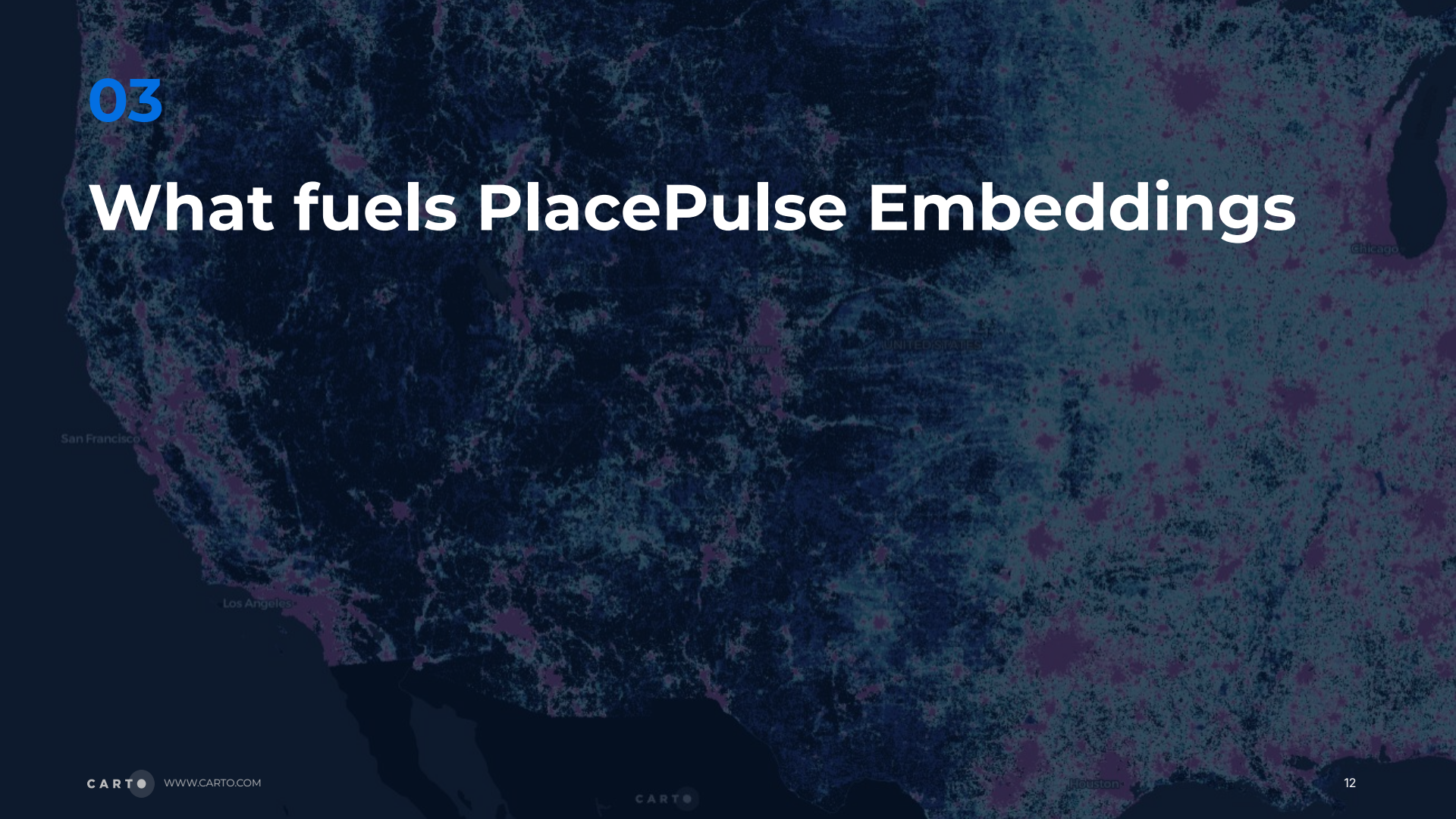
One representation for every populated location, on a uniform H3 grid.

Reusable across many tasks

Similarity search, clustering, prediction and more from one dataset.




Toronto
Chicago
New York
UNITED STATES
Denver
Washington
Los Angeles
Atlanta
Houston
Monterrey
Miami
CARTO


03
What fuels PlacePulse Embeddings
San Francisco
Los Angeles
Chicago
Houston

AGS Spatial Data Product

- *Current year sociodemographics estimates*
- *Non-Resident Population*
- *Business/Daytime Employees and Establishments*
- *Demographic and Business Dimensions*
- *Chronic Health Conditions and Risk Factors*
- *Climate And Weather Risk*
- *Journey to Work*
- *Disposable & Discretionary Income*



From subscription to analysis

01

Subscribe in the Data Observatory

The embeddings arrive as a ready-to-query dataset in your warehouse, no pipeline to build.

02

Analyze in CARTO Workflows

Drag-and-drop components run the vector math at national scale, in-warehouse.

03

Model with BigQuery or Snowflake ML

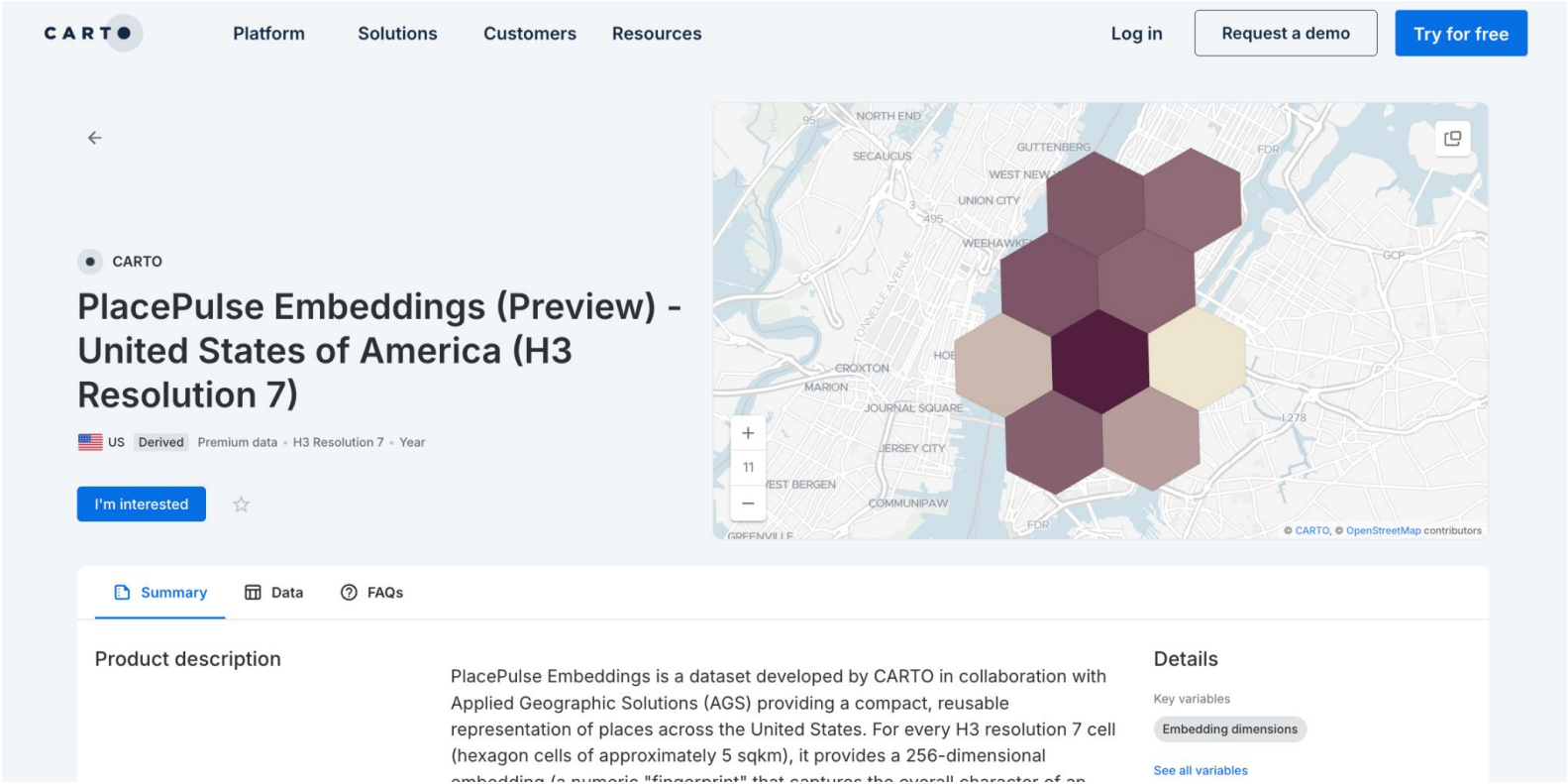
Feed the 256 dimensions into regression or classification pipelines.

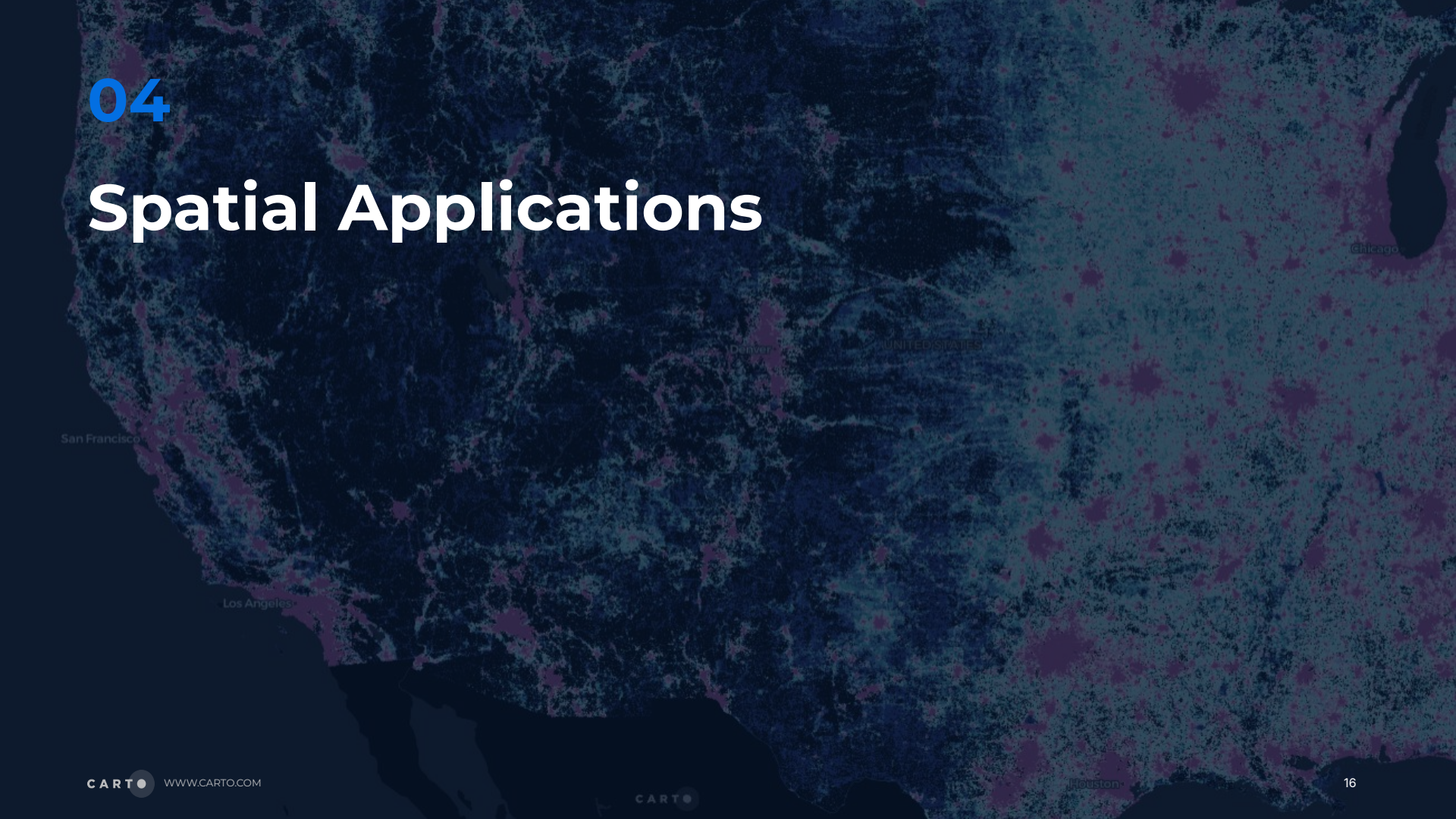
04

Bring your own data

Match your stores, customers or service areas to the embeddings that describe them.

Available in CARTO Data Catalog




04
Spatial Applications
San Francisco
Los Angeles
Denver
Chicago
Houston

Four ways to put it to work

Retrieve

Search large areas efficiently using cosine similarity by starting with a place you understand, then finding the places that resemble it.

e.g., Finding twin markets & expansion hotspots

Predict

Train a model where you have observations, then use it to estimate the same variable in places where it has never been measured.

e.g., Estimating revenue or foot traffic in new zones

Describe

Cluster places to reveal distinct place types, measure how typical or unusual each location is, or identify transitions and boundaries.

e.g., Neighborhood typology & anomaly detection

Reason

Use spatial AI systems to connect complex spatial characteristics, geographic context, and site insights directly to critical business questions.

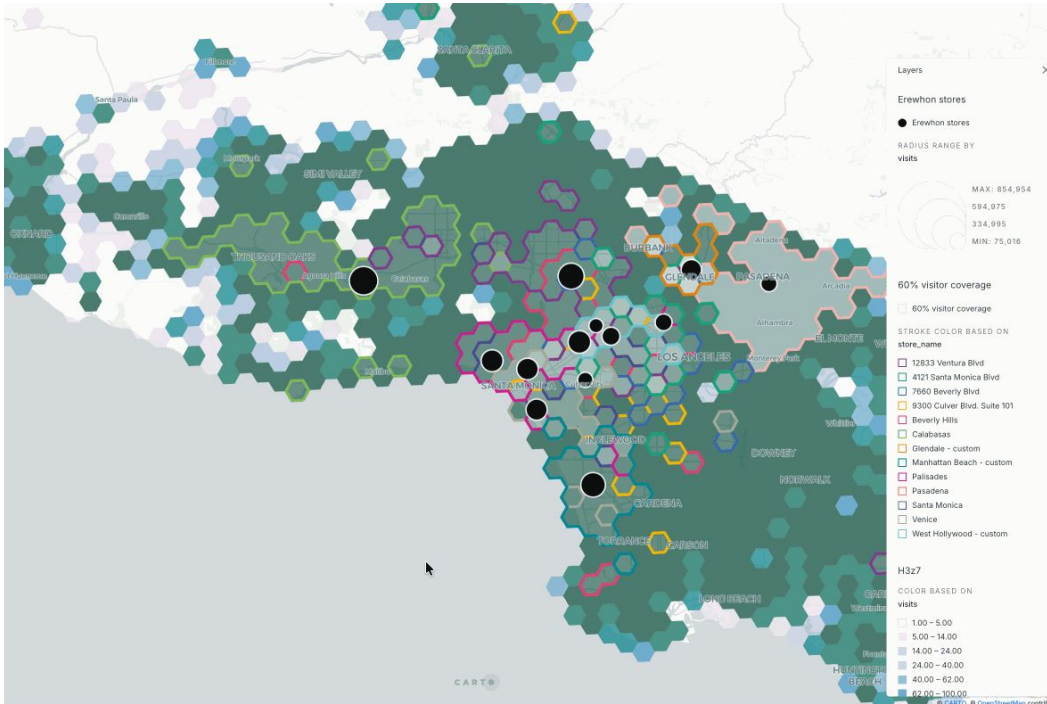
e.g., AI-driven site selection & spatial Q&A

Today's use case: Erewhon Expansion Analysis

Erewhon is an upscale, organic grocery store chain and pop-culture lifestyle brand based in LA. Its current 13 stores draw visitors from across the region.

This workshop uses ***visitor data (Advan)*** and ***PlacePulse embeddings*** to answer several business questions:

- *Where are Erewhon visitors coming from?*
- *Where should Erewhon target its next marketing campaign?*
- *Where should Erewhon open new stores?*



05

Hands on session



CARTO Workflows Toolkit

Analytics on Embeddings



Visualization

Visualize embeddings to reveal spatial patterns



Clustering

Group locations based on latent spatial patterns



Similarity Search

Find locations that behave alike in space



Change Detection

Detect spatial changes across temporal snapshots



Traditional ML

Bring embeddings into predictive models



Embedding Profile

Combines multiple embeddings into one optionally weighted profile vector



Mean Deviation

Find anomalies or the most representative locations



Neighbourhood Deviation

Highlight boundaries, gradients, and local heterogeneity



Spatial Aggregation

Translate embeddings to other geo-supports suitable for specific use cases



Vector Normalisation

Rescales vectors to unit length while preserving direction

CARTO MCP Server

CARTO inside any AI chat or AI app

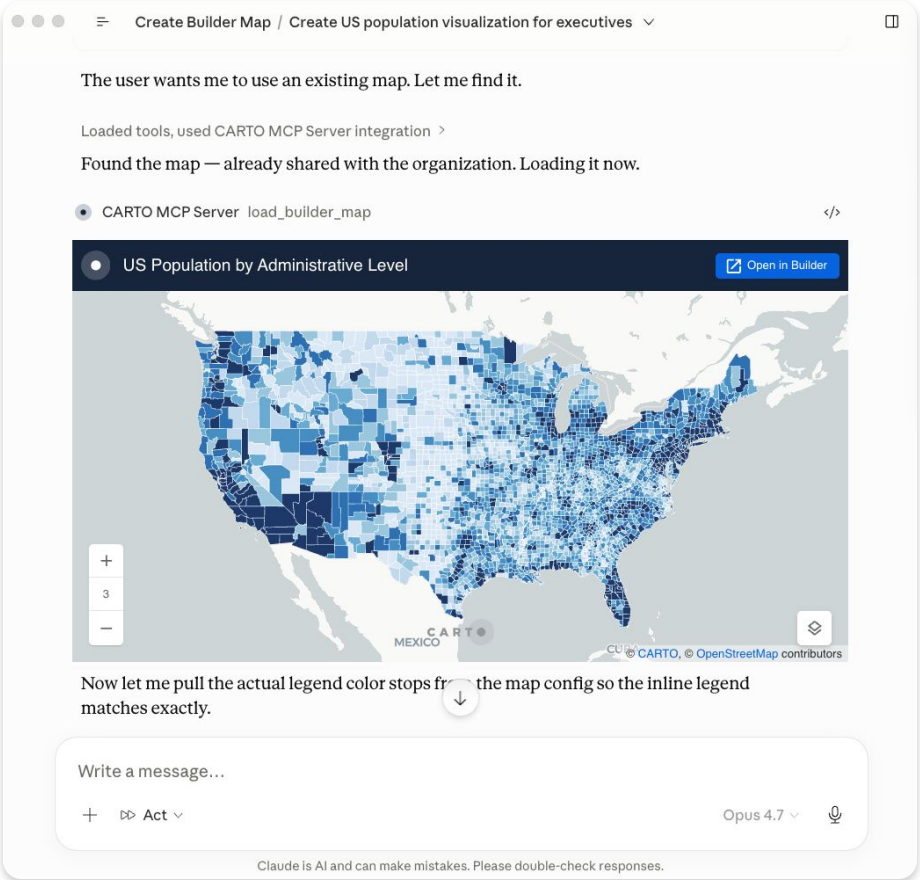
- → Explore data
- → Create & edit maps
- → Build & run workflows
- → Query & analyze
- → Import & export
- → Manage your workspace

Get started

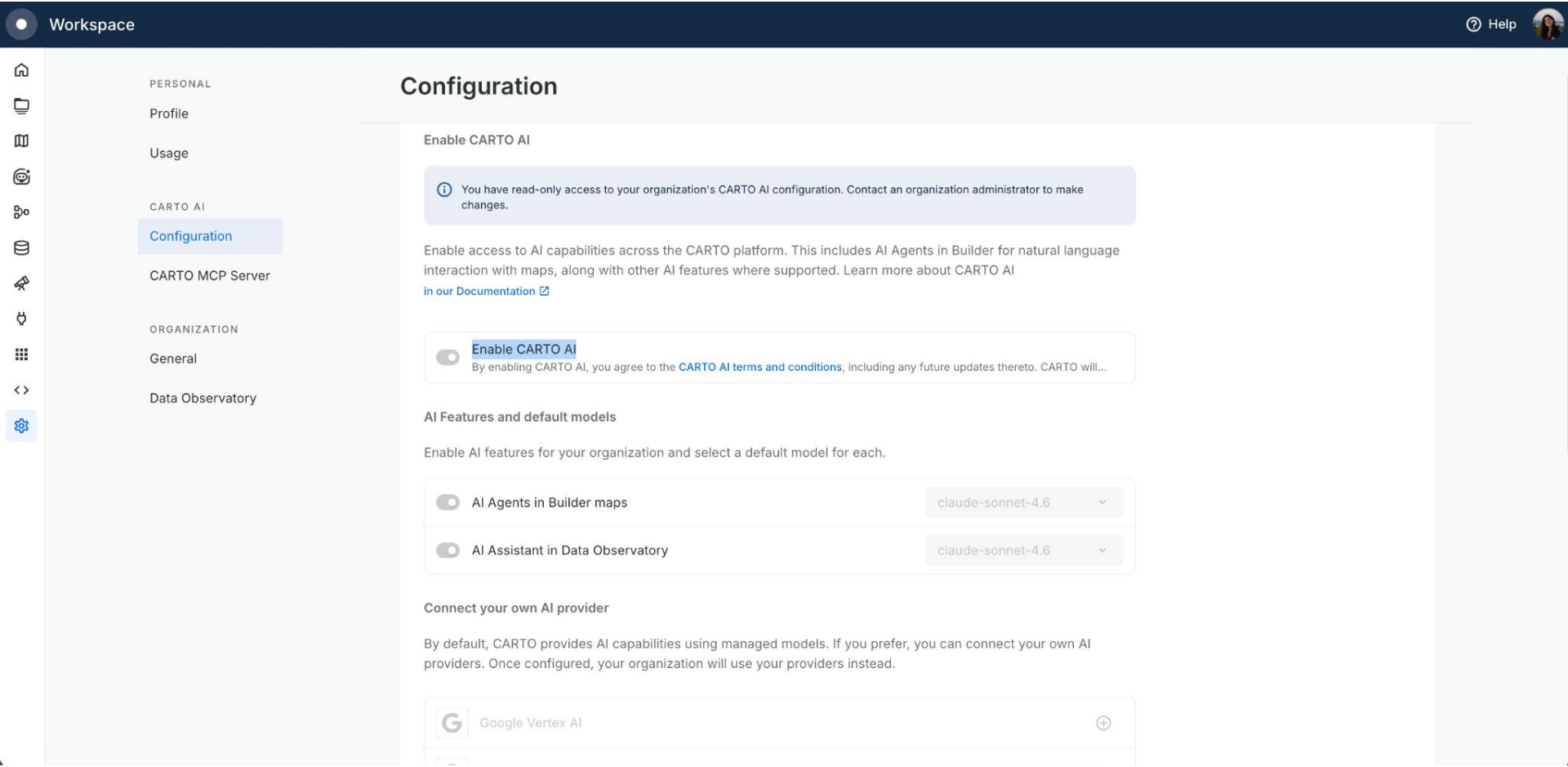
CARTO MCP Server CUSTOM

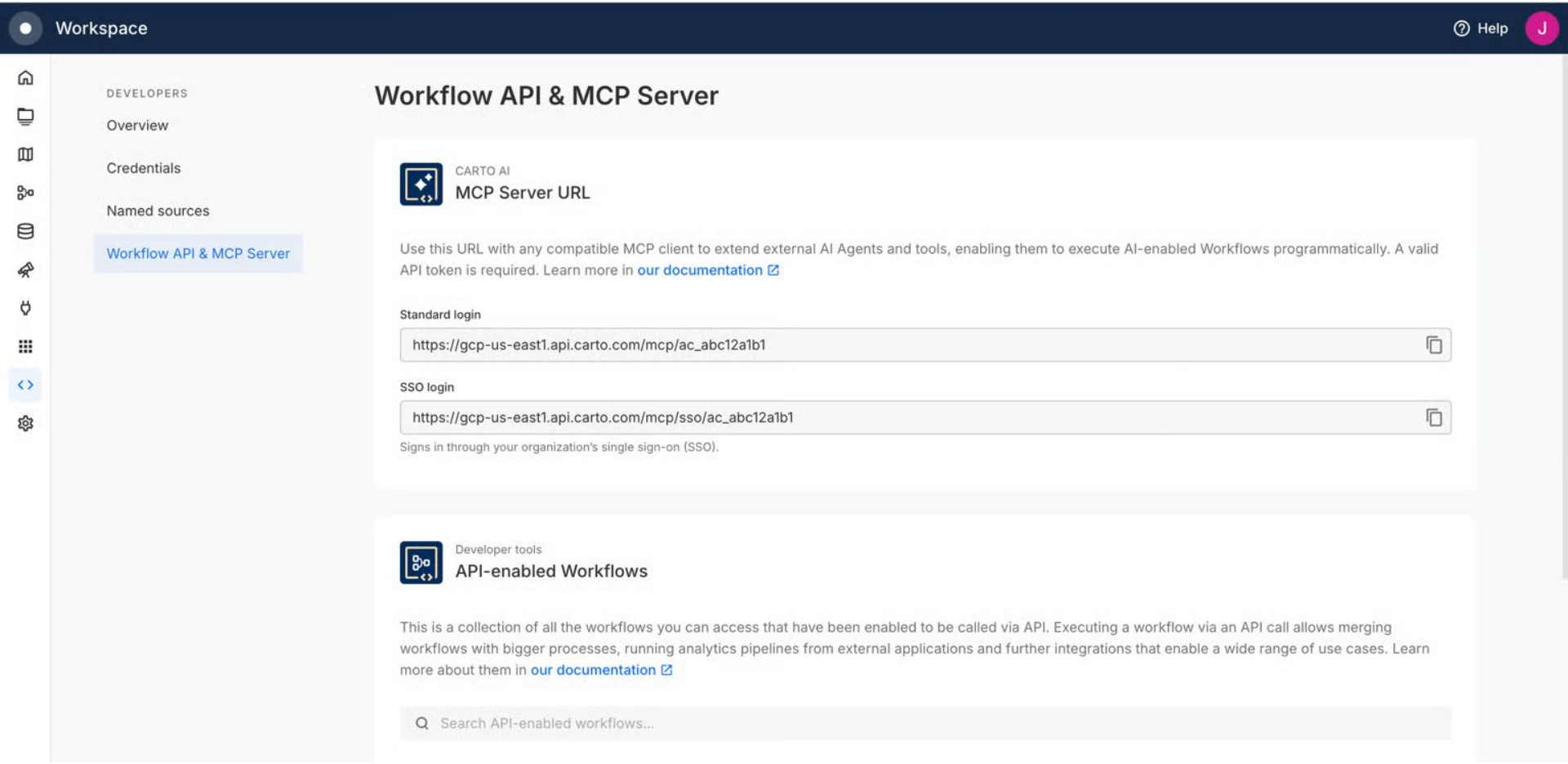
https://pinea.api.carto.com/mcp/acme

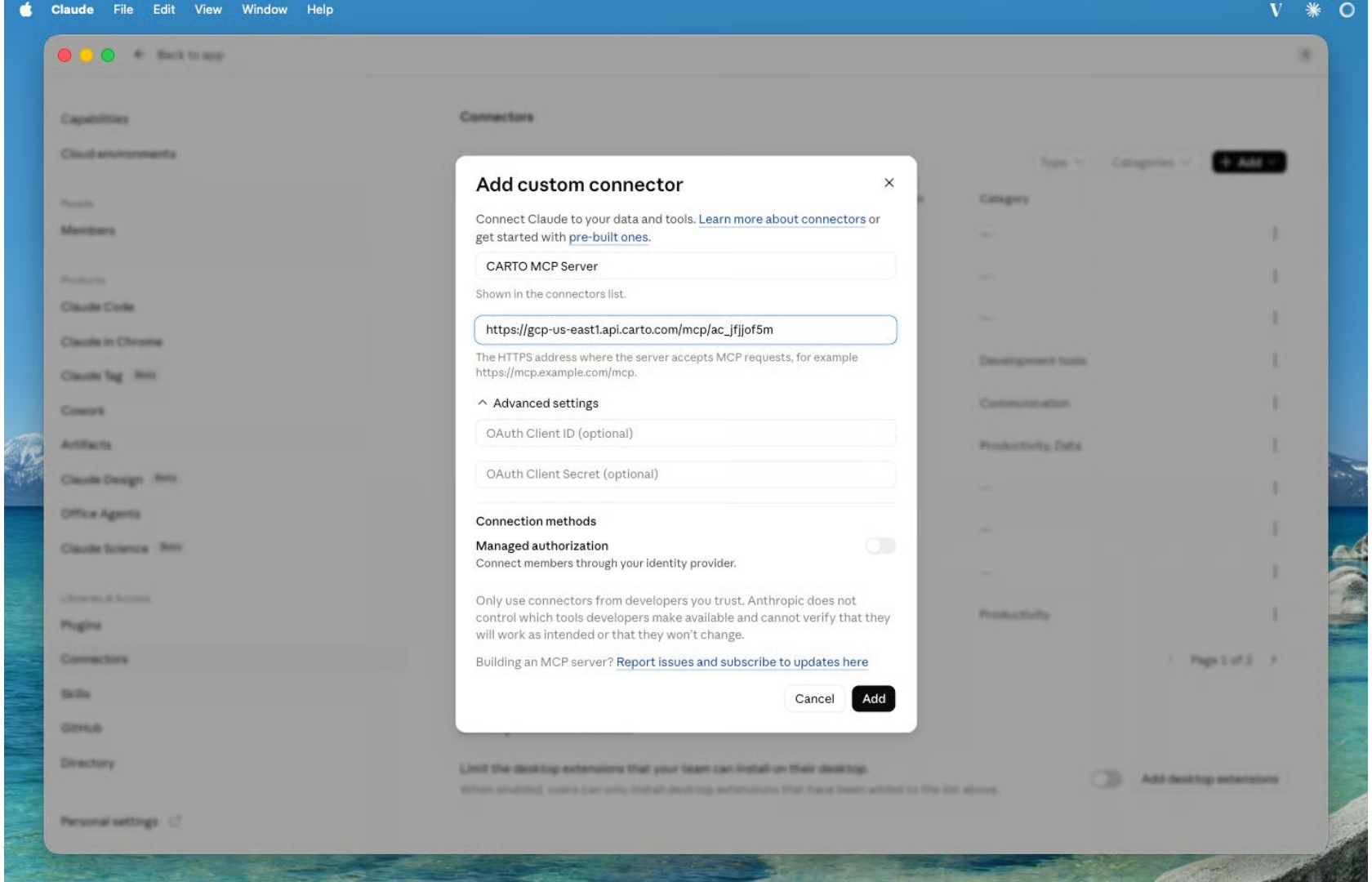
https://docs.carto.com/carto-for-agents/mcp-server



Let’s START!


Workspace
Help
PERSONAL
Profile
Usage
CARTO AI
Configuration
CARTO MCP Server
ORGANIZATION
General
Data Observatory
Configuration
Enable CARTO AI
You have read-only access to your organization's CARTO AI configuration. Contact an organization administrator to make changes.
Enable access to AI capabilities across the CARTO platform. This includes AI Agents in Builder for natural language interaction with maps, along with other AI features where supported. Learn more about CARTO AI in our Documentation
Enable CARTO AI
By enabling CARTO AI, you agree to the CARTO AI terms and conditions, including any future updates thereto. CARTO will...
AI Features and default models
Enable AI features for your organization and select a default model for each.
AI Agents in Builder maps
claude-sonnet-4.6
AI Assistant in Data Observatory
claude-sonnet-4.6
Connect your own AI provider
By default, CARTO provides AI capabilities using managed models. If you prefer, you can connect your own AI providers. Once configured, your organization will use your providers instead.
Google Vertex AI


Workspace
Help
J
DEVELOPERS
Overview
Credentials
Named sources
Workflow API & MCP Server
Workflow API & MCP Server
CARTO AI
MCP Server URL
Use this URL with any compatible MCP client to extend external AI Agents and tools, enabling them to execute AI-enabled Workflows programmatically. A valid API token is required. Learn more in our documentation
Standard login
https://gcp-us-east1.api.carto.com/mcp/ac_abc12a1b1
SSO login
https://gcp-us-east1.api.carto.com/mcp/sso/ac_abc12a1b1
Signs in through your organization's single sign-on (SSO).
Developer tools
API-enabled Workflows
This is a collection of all the workflows you can access that have been enabled to be called via API. Executing a workflow via an API call allows merging workflows with bigger processes, running analytics pipelines from external applications and further integrations that enable a wide range of use cases. Learn more about them in our documentation
Search API-enabled workflows...


Claude File Edit View Window Help
Add custom connector
Connect Claude to your data and tools. Learn more about connectors or get started with pre-built ones.
CARTO MCP Server
Shown in the connectors list.
https://gcp-us-east1.api.carto.com/mcp/ac_jfjjof5m
The HTTPS address where the server accepts MCP requests, for example https://mcp.example.com/mcp.
Advanced settings
OAuth Client ID (optional)
OAuth Client Secret (optional)
Connection methods
Managed authorization
Connect members through your identity provider.
Only use connectors from developers you trust. Anthropic does not control which tools developers make available and cannot verify that they will work as intended or that they won't change.
Building an MCP server? Report issues and subscribe to updates here
Cancel
Add

The prompts

Using CARTO MCP Server

Build a CARTO workflow that computes each Erewhon store's "60% trade area" (the smallest set of origin cells accounting for 60% of that store's visits).

SOURCE: `cartobq.docs.erewhon_visits_by_h3z7_la`

LOGIC, per store:

1. Total that store's visits across all its origin cells.

2. Rank its cells by visits descending; tie-break on h7 so it is reproducible.

3. Keep every cell up to AND INCLUDING the one that first crosses 60%.

Now, run the workflow and create a CARTO Builder map showing:

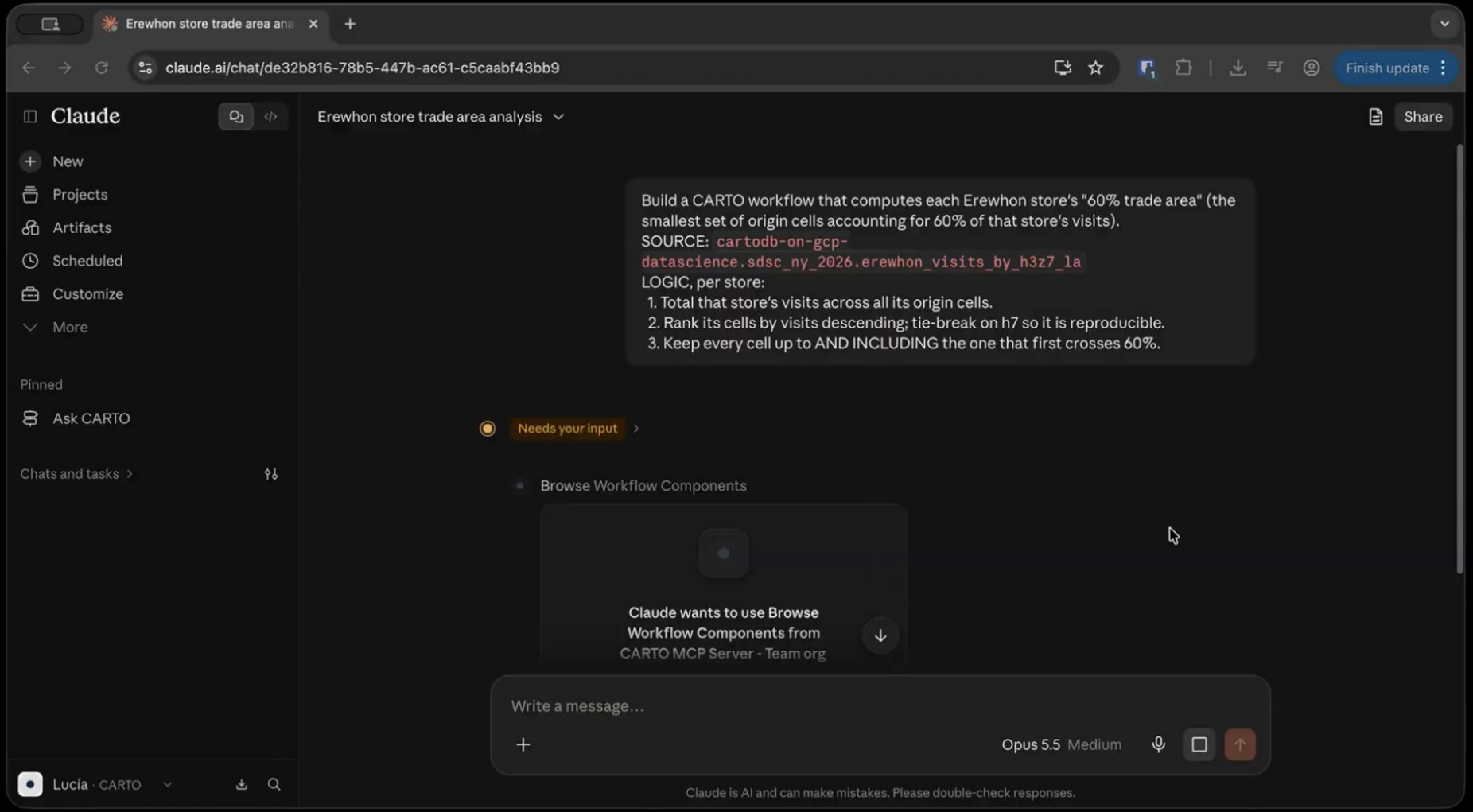
1. Erewhon stores – `erewhon_la_stores` (point layer).

2. 60% trade areas – the workflow output table (polygon layer).

3. Visits per origin cell – `erewhon_visits_by_h3z7_la` (H3 layer)

Add any relevant widgets that can help me analyze the results. I would especially like to be able to filter origins (H3 cells) by how far they are from the store they serve to.

Add a short map description explaining the map's layers and how to navigate through it.


Erewhon store trade area ana
claude.ai/chat/de32b816-78b5-447b-ac61-c5caabf43bb9
Finish update
Claude
Erewhon store trade area analysis
Share
New
Projects
Artifacts
Scheduled
Customize
More
Pinned
Ask CARTO
Chats and tasks
Build a CARTO workflow that computes each Erewhon store's "60% trade area" (the smallest set of origin cells accounting for 60% of that store's visits).
SOURCE: cartodb-on-gcp-datascience.sdsc_ny_2026.erewhon_visits_by_h3z7_la
LOGIC, per store:
1. Total that store's visits across all its origin cells.
2. Rank its cells by visits descending; tie-break on h7 so it is reproducible.
3. Keep every cell up to AND INCLUDING the one that first crosses 60%.
Needs your input
Browse Workflow Components
Claude wants to use Browse Workflow Components from CARTO MCP Server - Team org
Write a message...
Opus 5.5 Medium
Lucía · CARTO
Claude is AI and can make mistakes. Please double-check responses.

One step further...

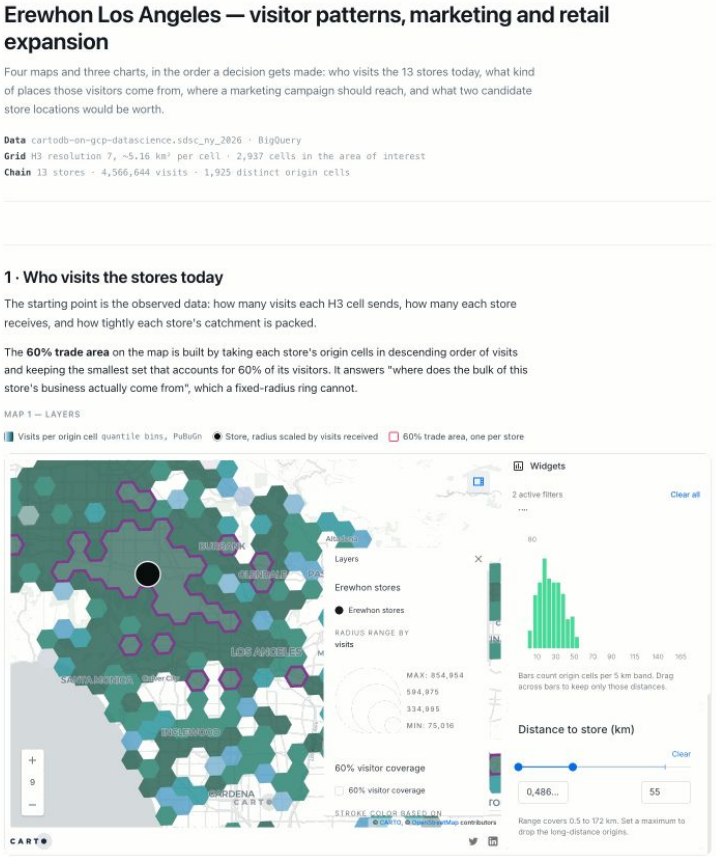
Using CARTO MCP Server

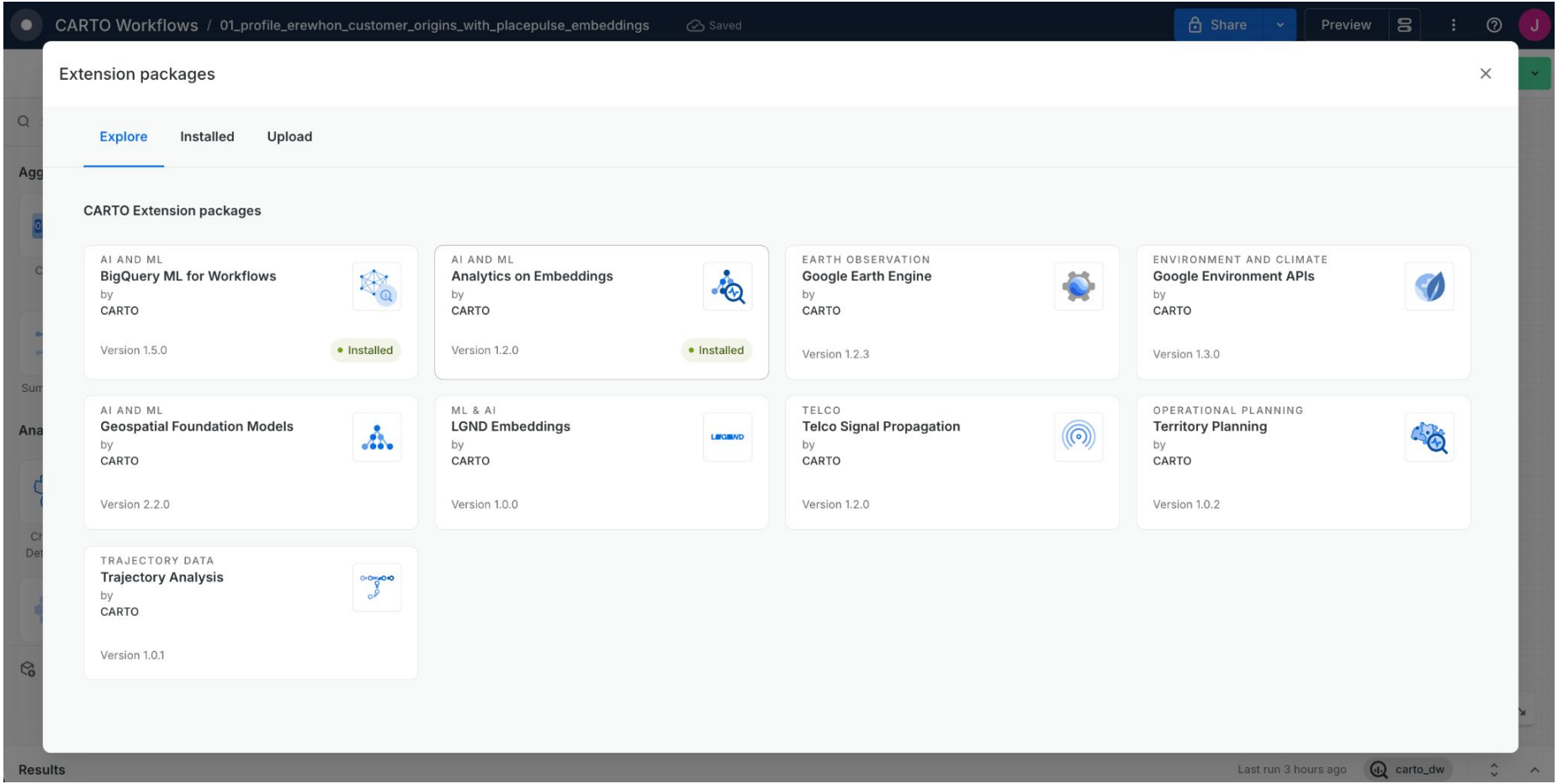
Build a full report with maps and graphs embedded

01 **Embedded Maps**
Integrate multiple interactive spatial visualizations seamlessly.

02 **Further Analysis**
Conduct visitor-weighted spatial profiling and data breakdown.

03 **Strategic Conclusions**
Synthesize metrics into clear actionable marketing insights.




CARTO Workflows / 01_profile_erewhon_customer_origins_with_placepulse_embeddings
Saved
Share
Preview
Extension packages
Explore
Installed
Upload
CARTO Extension packages
AI AND ML
BigQuery ML for Workflows
by
CARTO
Version 1.5.0
Installed
AI AND ML
Analytics on Embeddings
by
CARTO
Version 1.2.0
Installed
EARTH OBSERVATION
Google Earth Engine
by
CARTO
Version 1.2.3
ENVIRONMENT AND CLIMATE
Google Environment APIs
by
CARTO
Version 1.3.0
AI AND ML
Geospatial Foundation Models
by
CARTO
Version 2.2.0
ML & AI
LGND Embeddings
by
CARTO
Version 1.0.0
TELCO
Telco Signal Propagation
by
CARTO
Version 1.2.0
OPERATIONAL PLANNING
Territory Planning
by
CARTO
Version 1.0.2
TRAJECTORY DATA
Trajectory Analysis
by
CARTO
Version 1.0.1
Results
Last run 3 hours ago
carto_dw

Overview

Profile Erewhon customer origins with PlacePulse embeddings

This workflow uses PlacePulse embeddings for Los Angeles at H3 resolution 7. An embedding is a numeric vector that describes the structural character of one H3 cell.

- **Top branch:** maps the embeddings and measures how unusual each cell is.
- **Bottom branch:** groups the cells that send visitors to Erewhon stores into clusters.

Data

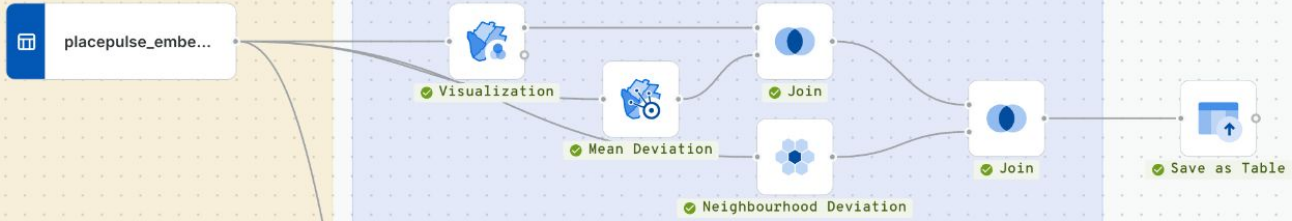
PlacePulse embeddings

One row per H3 cell in Los Angeles. `GEOID` is the H3 index. The `emb_*` columns are the embedding dimensions.

Spatial profiles

Describe the spatial profile of LA

1. **Visualization** reduces the embeddings to three PCA components and maps them to RGB. Similar colours mean similar cells.
2. **Mean Deviation** scores how far each cell is from the mean embedding of LA.
3. **Neighbourhood Deviation** compares each cell with its first ring of neighbours. High values mark boundaries between different areas.



Data

Erewhon visits

Visits from each H3 cell (`h7`) to each of the 13 Erewhon stores (`store_name` , `visits`).

Clustering

Cluster the Erewhon origin cells

1. **Join** keeps only the cells that send visitors to a store and attaches their embeddings.
2. **Drop Columns** and **Remove Duplicated** remove the store and visit fields, so each cell appears once.
3. **Clustering** runs K-means with k = 5 on the embeddings to group cells with a similar structural profile.



Overview

Target the marketing campaign

Erewhon is an upscale, organic grocery store chain and pop-culture lifestyle brand based in the Greater Los Angeles area. This workflow uses the visitors of its 13 stores to find the areas of LA where a marketing campaign has the most potential.

Each store gets a profile, which is the visit-weighted mean of the PlacePulse embeddings of the cells that its visitors come from. An embedding is a numeric vector that describes the urban character of one H3 cell.

- **Store profiles:** one unit-length profile vector for each store.
- **Store similarity:** compares the 13 profiles with each other.
- **Targeting score:** ranks each LA cell by its similarity to the profiles and by its current visits.

Data

Inputs

- **PlacePulse embeddings**: One row per H3 cell in Los Angeles. `GEOID` is the H3 index. The `emb_*` columns are the embedding dimensions.
- **Erewhon visits**: Visits from each H3 cell (`h7`) to each of the 13 Erewhon stores (`store_name` , `visits`).



Store profiles

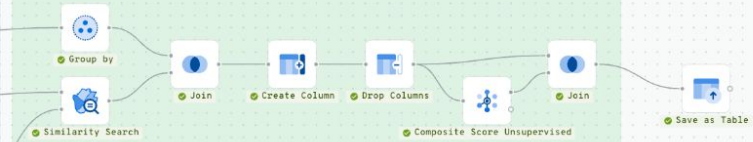
Build one profile for each store

1. **Join** attaches the embedding of each origin cell to its visits.
2. **Embedding Profile** calculates the visit-weighted mean embedding for each store. A cell with many visits pulls the profile more than a cell with few visits.
3. **Vector Normalisation** rescales each profile to length 1. With unit-length vectors, the Dot product metric returns the same values as Cosine.

Targeting score

Find similar cells with few visits

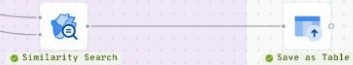
1. **Similarity Search** compares each LA cell with the 13 store profiles and takes the average similarity.
2. **Group by** sums the current visits of each cell across all stores.
3. **Join, Create Column** and **Drop Columns** attach the visits to every cell. A cell with no visits gets 0.
4. **Composite Score Unsupervised** combines the two variables with equal weights. The visits variable is reversed, so a cell with high similarity and few visits gets the highest score.



Store similarity

How alike are the store profiles?

Similarity Search compares each store profile with every other store profile.



Overview

Estimate visitor counts for new stores

A regression model learns the visits of the 13 current stores from two things: the PlacePulse embedding of each origin cell and its distance to the store. The model then predicts the visits for the new stores in Costa Mesa and Thousand Oaks.

- **Train:** fits the model on the current stores.
- **Prepare:** builds one row for each cell and new store.
- **Predict:** estimates the visits and the trade area of each new store.

Data

Inputs

- `erewhon_la_stores` : location of the 13 current stores.
- `erewhon_visits_by_h3z7_la` : visits from each H3 cell (`h7`) to each current store.
- `placepulse_embeddings_la` : one PlacePulse embedding for each H3 cell in LA (`GEOID`).
- `erewhon_new_stores` : location of the 2 new stores.

Train

Train the visitor model

1. **H3 Center** gets the centre point of each cell. **Join** attaches the embeddings to the visits, and the second **Join** attaches the store location.
2. **Create Column** calculates the distance from the cell to the store. A second **Create Column** calculates `log_visits` = LOG(visits + 1), which reduces the effect of cells with very many visits.
3. **Drop Columns** and **Generate UUID** keep the model inputs and add a unique `id` .
4. **Create Regression Model** trains a random forest regressor that predicts `log_visits` .

Prepare

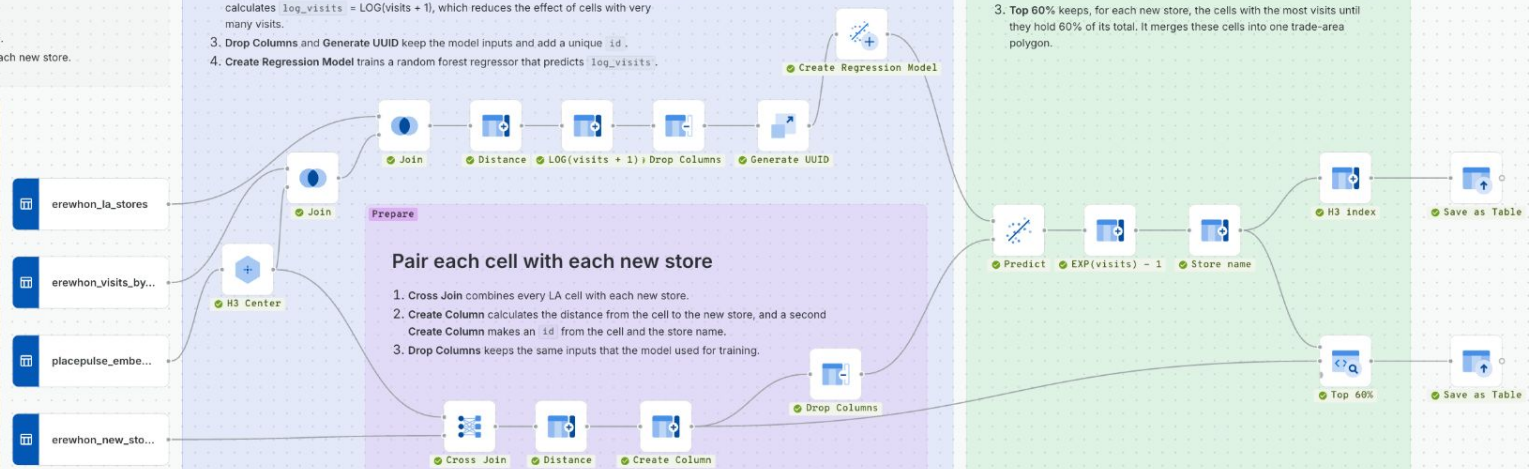
Pair each cell with each new store

1. **Cross Join** combines every LA cell with each new store.
2. **Create Column** calculates the distance from the cell to the new store, and a second **Create Column** makes an `id` from the cell and the store name.
3. **Drop Columns** keeps the same inputs that the model used for training.

Predict

Estimate visits and trade areas

1. **Predict** applies the model. **Create Column** converts the prediction back to visits with EXP(prediction) - 1.
2. Two **Create Column** steps split the `id` back into `store_name` and `h3` .
3. **Top 60%** keeps, for each new store, the cells with the most visits until they hold 60% of its total. It merges these cells into one trade-area polygon.



Want to know more about how to apply PlacePulse Embeddings to different use cases?

Take a look at our Blog!

Finding the next EV Charging Hotspots

Predicted infrastructure vs. reality

01

Predict the expected level

Model the charging infrastructure a neighbourhood should have from its profile.

02

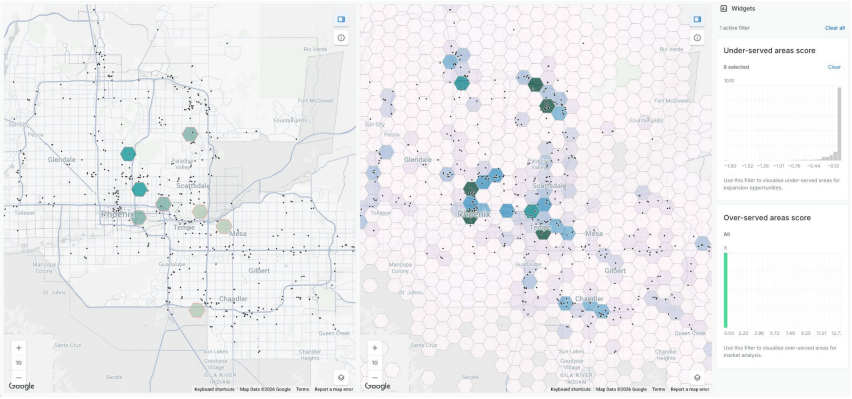
Compare to what's on the ground

The gap reveals areas primed for EV adoption but under-served today.

03

Get an expansion roadmap

Green cells flag opportunity; red cells flag saturation.



Sources | Components

Composite

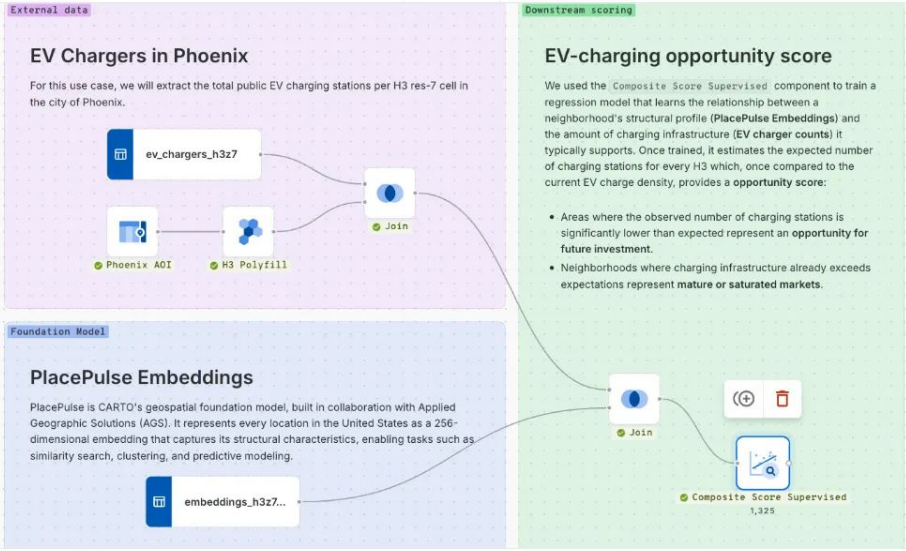
2 components by **"Composite"**

Statistics (2)

Composite Score Supervised

Composite Score Unsupervised

Manage Extension Packages



Run

STATISTICS

Composite Score Supervised

SUCCESS

This component derives a spatial composite score as the residuals of a regression model which is used to detect areas of under- and...

Composite Score Supervised documentation

Settings

Column with unique geographic identifier

GEOID

Input variables — Remove all

- CHARGER_COUNT
- emb_0
- emb_1
- emb_2
- emb_3
- emb_4
- emb_5
- emb_6
- emb_7
- emb_8
- emb_9
- emb_10

Results | Messages | Data | Map | SQL

Last run 36 min. ago — bq-ds-conn

Composite Score Supervised — 1,325 ROWS • 2 COLS.

spatial_score number	GEOID string
-1.497	8729b6d14ffffff
-1.243	8748eba36ffffff
-1.176	8748eba48ffffff
-1.137	8729b6d05ffffff

Rows per page 50 — 1-50 of 1325

Exploring where could a brand grow

Whole Foods expansion example

01

Build a retail fingerprint

Best-performing stores share a consistent embedding signature; weak ones are outliers.

02

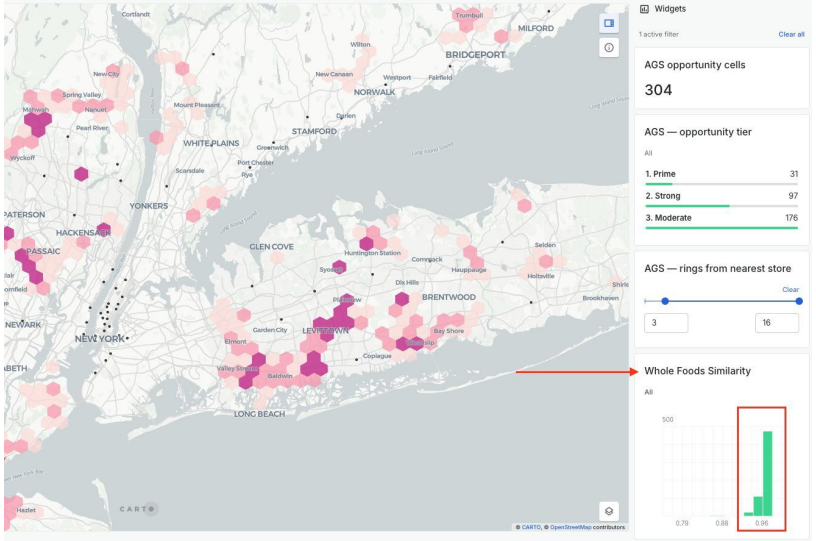
Score every trade area in the US

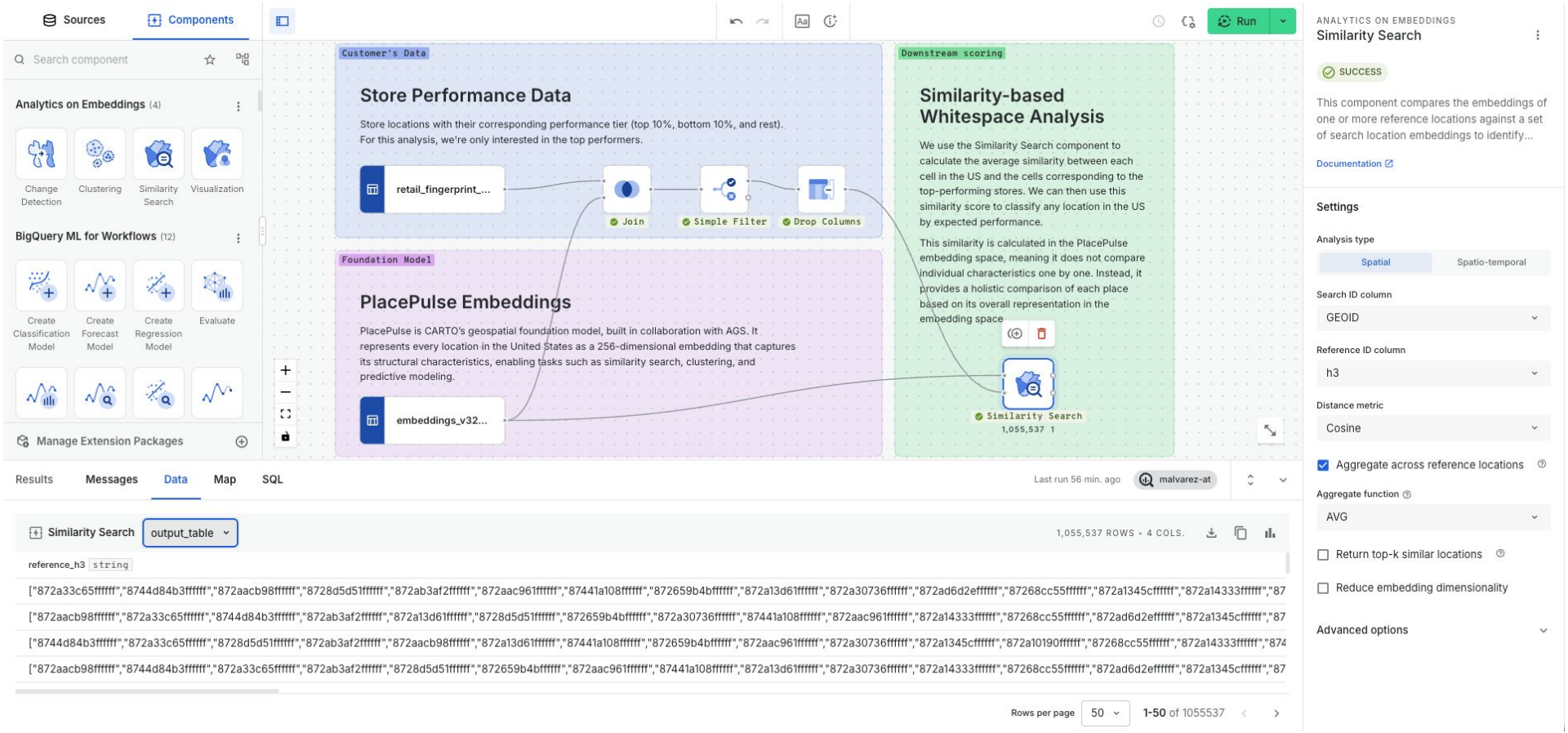
Each location is ranked by how closely it matches that fingerprint.

03

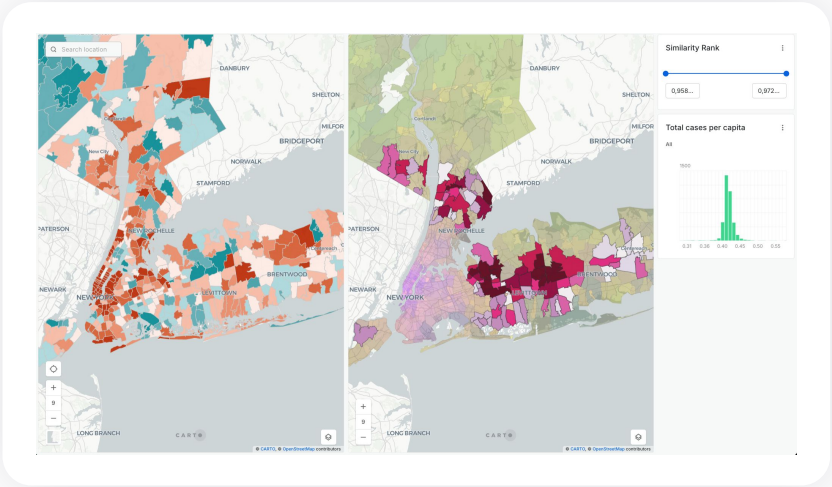
Prioritize by opportunity tier

A nationwide ranking grouped from Prime to Low Potential.




Sources
Components
Search component
Analytics on Embeddings (4)
Change Detection
Clustering
Similarity Search
Visualization
BigQuery ML for Workflows (12)
Create Classification Model
Create Forecast Model
Create Regression Model
Evaluate
Manage Extension Packages
Run
Customer's Data
Store Performance Data
Store locations with their corresponding performance tier (top 10%, bottom 10%, and rest). For this analysis, we're only interested in the top performers.
retail_fingerprint_...
Join
Simple Filter
Drop Columns
Foundation Model
PlacePulse Embeddings
PlacePulse is CARTO's geospatial foundation model, built in collaboration with AGS. It represents every location in the United States as a 256-dimensional embedding that captures its structural characteristics, enabling tasks such as similarity search, clustering, and predictive modeling.
embeddings_v32...
Downstream scoring
Similarity-based Whitespace Analysis
We use the Similarity Search component to calculate the average similarity between each cell in the US and the cells corresponding to the top-performing stores. We can then use this similarity score to classify any location in the US by expected performance.
This similarity is calculated in the PlacePulse embedding space, meaning it does not compare individual characteristics one by one. Instead, it provides a holistic comparison of each place based on its overall representation in the embedding space
Similarity Search
1,055,537 1
Results
Messages
Data
Map
SQL
Last run 56 min. ago
malvarez-at
Similarity Search
output_table
1,055,537 ROWS • 4 COLS.
reference_h3 string
["872a33c65fffffff","8744d84b3fffffff","872aacb98fffffff","8728d5d51fffffff","872ab3af2fffffff","872aac961fffffff","87441a108fffffff","872659b4bfffffff","872a13d61fffffff","872a30736fffffff","872ad6d2efffffff","87268cc55fffffff","872a1345cfffffff","872a14333fffffff","87
["872aacb98fffffff","872a33c65fffffff","8744d84b3fffffff","872ab3af2fffffff","872a13d61fffffff","8728d5d51fffffff","872659b4bfffffff","872a30736fffffff","87441a108fffffff","872aac961fffffff","872a14333fffffff","87268cc55fffffff","872ad6d2efffffff","872a1345cfffffff","87
["8744d84b3fffffff","872a33c65fffffff","8728d5d51fffffff","872ab3af2fffffff","872aacb98fffffff","872a13d61fffffff","87441a108fffffff","872659b4bfffffff","872aac961fffffff","872a30736fffffff","872a1345cfffffff","872a10190fffffff","87268cc55fffffff","872a14333fffffff","874
["872aacb98fffffff","8744d84b3fffffff","872a33c65fffffff","872ab3af2fffffff","8728d5d51fffffff","872659b4bfffffff","872aac961fffffff","87441a108fffffff","872a13d61fffffff","872a30736fffffff","872a14333fffffff","87268cc55fffffff","872ad6d2efffffff","872a1345cfffffff","87
Rows per page
50
1-50 of 1055537
ANALYTICS ON EMBEDDINGS
Similarity Search
SUCCESS
This component compares the embeddings of one or more reference locations against a set of search location embeddings to identify...
Documentation
Settings
Analysis type
Spatial
Spatio-temporal
Search ID column
GEOID
Reference ID column
h3
Distance metric
Cosine
Aggregate across reference locations
Aggregate function
AVG
Return top-k similar locations
Reduce embedding dimensionality
Advanced options

Exploring new insurance markets



Life & health insurance example

01

Start from anchor markets

Take the ZIPs where the strongest policyholders already cluster.

02

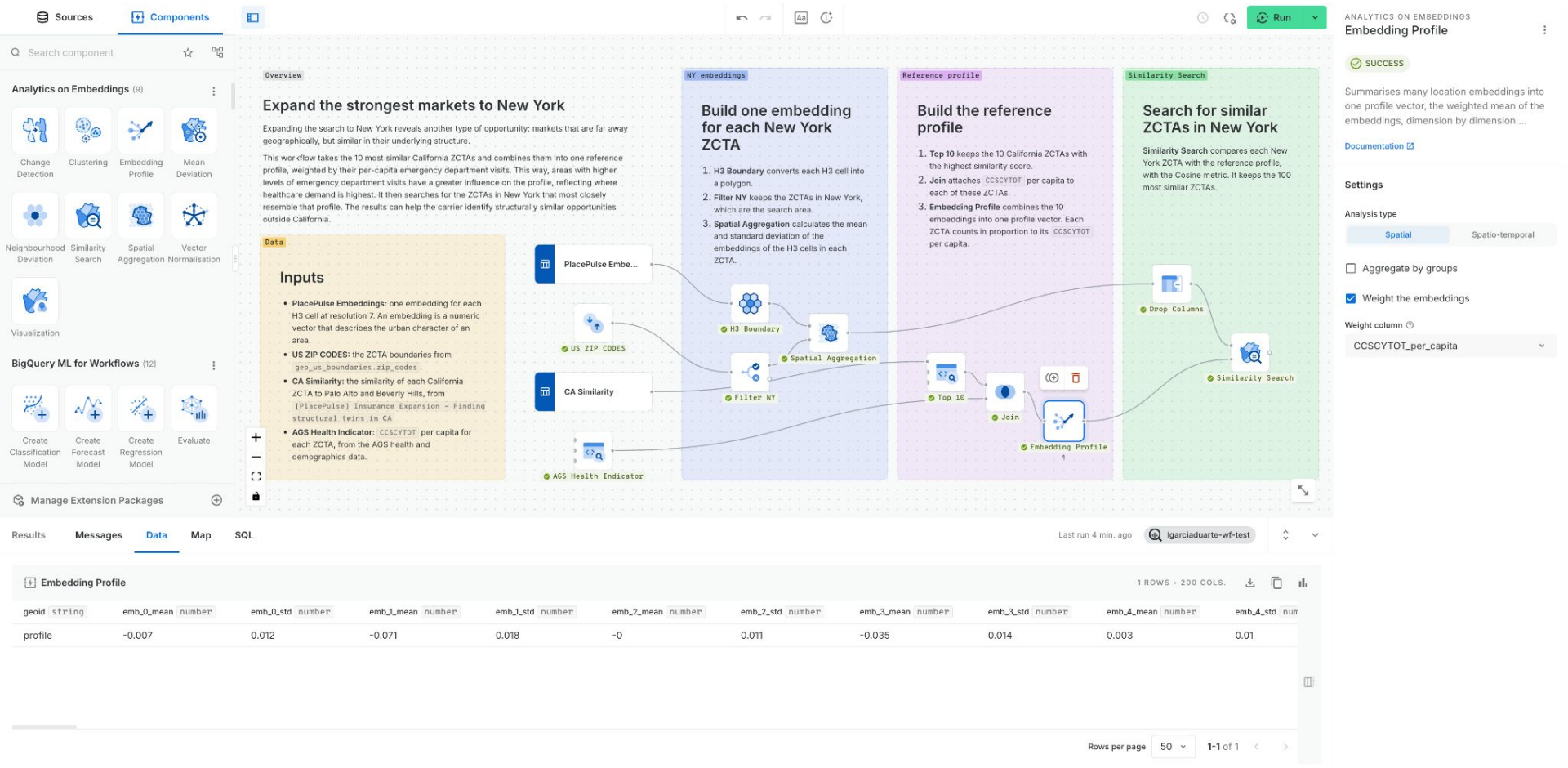
Search for structural twins

Surface look-alike markets in-state to deepen, and out-of-state to expand.

03

Layer in health context

Overlay chronic-condition burden to refine positioning by market.



ANALYTICS ON EMBEDDINGS

Embedding Profile

SUCCESS

Summarises many location embeddings into one profile vector, the weighted mean of the embeddings, dimension by dimension....

Documentation

Settings

Analysis type: Spatial | Spatio-temporal

- [ ] Aggregate by groups
- [x] Weight the embeddings

Weight column: CCSCYTOT_per_capita

Results | Messages | Data | Map | SQL

Last run 4 min. ago — lgarciaduarte-wf-test

Embedding Profile — 1 ROWS · 200 COLS.

geoid string	emb_0_mean number	emb_0_std number	emb_1_mean number	emb_1_std number	emb_2_mean number	emb_2_std number	emb_3_mean number	emb_3_std number	emb_4_mean number	emb_4_std num
profile	-0.007	0.012	-0.071	0.018	-0	0.011	-0.035	0.014	0.003	0.01

Rows per page 50 — 1-1 of 1

06

PlacePulse Takeaways

The feature engineering is already done

2,288 attributes arrive as one 256-dimension vector for about a million H3 cells. The work moves from assembling variables to asking the question.

One dataset answers many questions

Similarity search, clustering, change detection, and predictive models all run on the same vectors. Retail, insurance, and EV siting used one subscription today.

The analysis runs where your data already is

Subscribe through the Data Observatory and the vector math executes in BigQuery or Snowflake, so the data never moves and the audit trail stays yours.

Thanks!

Lucía García - Duarte
Senior Data Scientist @ CARTO
lgarciaduarte@cartodb.com
/luciagds/

Gary Menger
President @ Applied Geographic Solutions
gary@appliedgeographic.com
/gary-menger-61658513/

Spatial Data Science Conference 2026, New York